

Figure 1: Main window with file browser, console, variable browser and command history

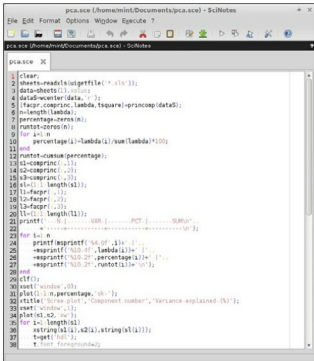


Figure 2: SciNotes editor

the screenshots are under Linux Mint 16 Xfce. Xcos and the relationship between Scilab and other languages will perhaps be explored in another article in the future.

Data analysis, in a nutshell

Let's consider a table from the book 'Introduzione alla Chemiometria' by Todeschini (see References), in which the rows represent the wines (38 objects, numbered from 1 to 38) and the columns are the chemical elements (17 variables, numbered from 1 to 17). In particular, the chemical elements are: Cd, Mo, Mn, Ni, Cu, Al, Ba, Cr, Sr, Pb, B, Mg, Si, Na, Ca, P and K. I can represent these wines on a 17-dimensional plot; each wine is a single point in this space. In Hierarchical Cluster Analysis (HCA), the distances between objects (or

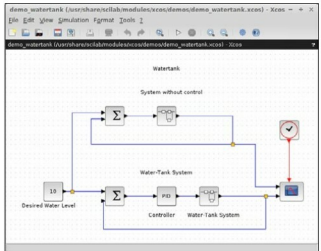


Figure 3: Xcos (from the demos)

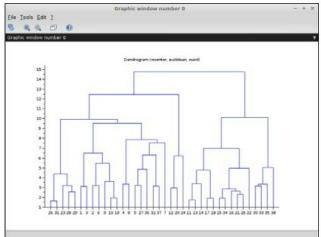


Figure 4: Dendrogram

variables) are calculated and then plotted in a tree-shaped distance map called a dendrogram. In principal component analysis (PCA), linear combinations of the variables are calculated. This transformation does not change the relationships between the objects. In this new space, the axes are orthogonal (completely uncorrelated) to one another and ordered by their importance, according to the variance that they explain. For this reason, it's possible to look for relationships among the objects in a few-dimensions plot.

How to go about it

For the hierarchical cluster analysis (the IPT3 module must be installed), it's first necessary to read the data table via `readxls`. Then, a data scaling is carried out: `center` is for the column centering (each column has a mean = 0) and `wcenter` is for the column standardisation (each column has a mean = 0 and standard deviation = 1). `pdist` calculates the distance between the objects. The available methods to calculate the distance are listed below:

- che (chebyshev);